CO4-Assessement Tool-01

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Course: Computer architecture

1. Mobile Device – Cache and DRAM

Analysis:

- L1 Cache: Very small, fastest memory; stores frequently used instructions/data.
- L2 Cache: Larger than L1 but slightly slower; stores data not found in L1.
- L3 Cache: Larger and slower; shared by CPU cores and useful during app switching.
- DRAM: Large main memory but much slower than cache; stores active applications and their data.

Proposed Hierarchy:
CPU → L1 → L2 → L3 → DRAM → Storage

This hierarchy gives high hit rate and bandwidth while keeping latency low. Frequently used application data remains in L1/L2/L3, reducing DRAM accesses and improving application-switching speed.

2. Cloud Database – Cache and Virtual Memory

Analysis:
Cache reduces processor-memory latency by keeping frequently accessed instructions and database data close to the CPU. Virtual memory allows programs to use more memory than physical DRAM, but page faults cause delays because data must be loaded from storage.

Recommended Improvements:

- Use larger/faster L1, L2 and L3 caches.
- Use high-bandwidth DDR5 DRAM.
- Increase physical RAM to reduce paging.
- Use SSD/NVMe instead of HDD for faster page access.
- Use database buffer/cache pools.
- Apply efficient memory allocation and page-replacement methods.
- Reduce unnecessary page faults through proper workload management.

Result: Fewer cache misses and page faults → lower access latency → higher database throughput.

3. Autonomous Vehicle – SRAM, DRAM and MRAM

Memory	Latency	Capacity	Volatile	Main Use
SRAM	Very low	Small	Yes	CPU cache
DRAM	Low	Large	Yes	Sensor-data processing
MRAM	Low	Medium	No	Persistent/critical data

Proposed Hierarchy:
CPU → SRAM → DRAM → MRAM → Storage

- SRAM handles real-time, frequently accessed sensor data.
- High-bandwidth DRAM stores large camera, LiDAR and radar data.
- MRAM provides non-volatile storage for important parameters and data.
- Use high-bandwidth DRAM and parallel memory channels to support continuous sensor streams.

Result: High bandwidth, low latency and reliable real-time sensor processing.

4. Edge AI Device – NAND Flash, DRAM and RRAM

Memory	Speed	Capacity	Persistence	Use
NAND Flash	Medium/Slow	High	Non-volatile	Model storage
DRAM	Very fast	Medium	Volatile	AI processing
RRAM	Fast	Medium	Non-volatile	Fast model/data storage

Proposed Hierarchy:
AI Processor → DRAM → RRAM → NAND Flash

- NAND Flash stores large AI models permanently.
- RRAM provides faster persistent model access.
- DRAM stores model parameters, intermediate results and temporary data.
- Frequently used models can be kept in RRAM/DRAM to reduce loading time.

Result: Faster model loading, lower energy consumption and improved AI inference performance.

5. Gaming Console – L3 Cache and DRAM

Analysis:

- L3 Cache: Small compared with DRAM but much faster; stores frequently accessed game data.
- DRAM: Much larger and provides high bandwidth for textures, audio, maps and other game assets.
- When data is found in L3, access is fast. A cache miss requires fetching data from DRAM, increasing latency.

Proposed Improvements:

- Increase L3 cache capacity.
- Use high-bandwidth GDDR6/GDDR6X or similar memory.
- Increase memory channels/bandwidth.
- Use intelligent prefetching for upcoming game assets.
- Keep frequently used textures and game data cached.
- Use fast SSD storage for large asset loading.

Result: Faster data transfer → fewer stalls → reduced frame drops → smoother gaming during large scene transitions.